Wifaq at NADI 2026 Shared Task: Data Expansion and System Combination for Robust Arabic ASR

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Abstract

Wifaq is our entry to NADI 2026 Subtask 1.1, which evaluates country-level Arabic automatic speech recognition across eight dialects under noisy conditions. We fully fine-tune four 600M-parameter W2v-BERT 2.0 CTC models on nested mixtures of task and external speech, decode them with a shared 5-gram KenLM, and combine their hypotheses by word voting; each run fits on one 16 GB NVIDIA RTX 4070 Ti SUPER. The submitted four-way ensemble ranked first among nine entries on the blind test with 43.23% WER and 20.0% CER. On a reused 800-utterance local holdout, it lowers WER from 46.12% for the strongest single model to 44.0%.

1 Introduction

Arabic speech recognition gets harder outside clean, standardized speech. Building on the first multidialectal Arabic speech shared task at NADI 2025 (Talafha et al., 2025), NADI 2026 evaluates systems against dialect shifts, spelling variation, code-switching, and poor recordings (Sullivan et al., 2026).

Subtask 1.1 covers Algerian, Egyptian, Jordanian, Mauritanian, Moroccan, Palestinian, Emirati, and Yemeni speech. Each test item has one country label and requires a normalized transcript.

Our main design question was how much complementary training data could add when every recognizer shared the same architecture and decoder. We therefore varied the acoustic training pool while holding the rest of the recognition pipeline fixed.

The submitted entry uses W2v-BERT 2.0 CTC models (Chung et al., 2021; Barrault et al., 2023), a shared 5-gram KenLM (Heafield, 2011), and a four-way ROVER-style ensemble (Fiscus, 1997). We document the nested training pools, decoding settings, and exact voting procedure. The four-way combination improves on the strongest single model by about 2.1 absolute WER points. We also report dialect-level errors and a distillation experiment that did not improve the system.

Code is available at `https://github.com/Alkurbi/nadi-2026-arabic-asr`. The repository excludes task data, external corpora, model weights, and submitted predictions.

2 Data

2.1 Task data and split

The organizers released 12,600 training utterances and 10,808 development utterances. The blind test set contains 4,000 utterances, with 500 from each of the eight dialects.

The training split is balanced at 1,575 utterances per dialect. Development contains 726 Algerian, 1,600 Egyptian, 1,599 Jordanian, 1,600 Mauritanian, 1,600 Moroccan, 900 Palestinian, 1,600 Emirati, and 1,183 Yemeni utterances.

We set aside 800 development utterances as a local holdout, sampling 100 per dialect, and folded the remaining 10,008 into the final acoustic-model training pool. The released list in `docs/holdout_ids.txt` identifies the exact holdout items.

This is not an untouched validation set. We reused it for checkpoint, language-model, and ensemble selection, so its scores may overstate generalization.

2.2 External speech

To broaden the training pool, we added 13,635 Casablanca utterances, a 2,814-utterance Omnilingual ASR subset, and 7,474 selected MoulSot utterances. Table 1 gives the exact composition.

Casablanca covers the same eight national dialects and provides transcriptions of diverse speech (Talafha et al., 2024). The Omnilingual subset uses the training splits of the `arq`, `aao`, `ary`, `arz`, `aeb`, `ayl`, `afb`, `acw`, `ars`, `acm`, `apc`, and `aec`

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Training source	Utterances
NADI training	12,600
Casablanca	13,635
Omnilingual subset	2,814
MoulSot subset	7,474
NADI development, folded in	10,008
Final training pool	46,531
Local holdout	800

Table 1: Acoustic training data for the strongest final model. The holdout has 100 utterances per dialect and is excluded from training.

Arabic configurations (Keren et al., 2025); we retained nonempty normalized transcripts between 0.3 and 18 seconds. MoulSot contributes automatically transcribed Moroccan Darija from several online genres (Atlasia, 2026). Its subset takes every fourth row of the 29,895-item local manifest, yielding 7,474 utterances and 9.41 hours.

We compared SHA-256 hashes for all 13,635 Casablanca audio files with 27,608 NADI training-side, development, and blind-test files and found no byte-identical cross-corpus audio. After official normalization, Casablanca shares exact transcripts with 48 of the 800 holdout utterances and 498 of the 10,008 folded development utterances, mostly short repeated phrases. Compatible speaker identifiers are unavailable, so speaker overlap could not be tested. The program chairs separately confirmed that Casablanca does not overlap with the Task 1.1 blind test and was permitted as training data.

2.3 Preprocessing

We convert audio to mono and process it at 16 kHz. Every model uses features produced by the W2v-BERT feature extractor.

The official normalization recipe is applied to training targets, language-model text, hypotheses, and references. It removes Arabic diacritics and tatweel, maps alef variants to bare alef, converts Eastern Arabic digits to Western digits, removes punctuation except %, and collapses whitespace.

We retain Arabic letters, Latin letters, and digits. The character tokenizer contains 101 entries, including the CTC blank, unknown, and word-delimiter symbols.

3 System

3.1 Acoustic model

Our starting point is `facebook/w2v-bert-2.0`, a 24-layer Conformer speech encoder with roughly 600M parameters (Chung et al., 2021; Barrault et al., 2023). The released encoder was pretrained on multilingual unlabeled speech. We add one adapter layer and a randomly initialized 101-way CTC projection (Graves et al., 2006).

All 605.8M parameters are fine-tuned. Training uses bfloat16, gradient checkpointing, AdamW, a peak learning rate of 10^{-4}, 300 warmup updates, linear decay, and seven epochs.

To make full fine-tuning fit in 16 GB, we sort examples by duration and pack them until the padded cost $B \cdot \max_i d_i$ reaches 45 seconds, where B is batch size and d_i is utterance duration. This limits wasted padding and keeps activation memory predictable on long clips. The final pool produces 5,331 dynamic batches per epoch.

We evaluate every 1,000 updates and select checkpoints by holdout WER. One run takes about six hours on an RTX 4070 Ti SUPER under Windows 11 and Python 3.11.

The implementation uses PyTorch 2.6, Transformers 4.57.6 (Wolf et al., 2020), Datasets 5.0.0, pyctcdecode 0.5.0, and jiwer 4.0.0.

3.2 Language-model decoding

The 5-gram KenLM (Heafield, 2011) is trained on exactly 519,198 normalized lines. A 26,435-line speech corpus, containing 12,600 NADI training transcripts, a separate 200-utterance training-side checkpoint split, and 13,635 Casablanca transcripts, is repeated three times for domain weighting. The remaining 439,893 lines are normalized posts of at least two words from the training split of the Arabic Dialect Identification dataset (Rezk, 2026). NADI development and holdout transcripts were not source corpora for the language model.

We decode each acoustic model with CTC beam search of width 150. The language-model weight is 0.5 and the word-insertion bonus is 1.5, selected on Egyptian development speech.

All four recognizers share this decoder and the same normalization. Their differences therefore come mainly from acoustic training data and independent optimization trajectories, rather than separate decoder tuning.

Split	System	WER	CER	n
Full dev	NADI-only AM, greedy	58.3	22.8	10,808
Full dev	NADI-only AM + LM	55.4	23.2	10,808
Holdout	*Checkpoint-26K*	48.7	20.9	800
Holdout	*Checkpoint-29K*	48.0	20.7	800
Holdout	*Checkpoint-37K*	47.6	20.5	800
Holdout	*Checkpoint-47K*	46.1	**19.4**	800
Holdout	Three-way vote	44.6	19.5	800
Holdout	Submitted 4-way vote	**44.0**	19.7	800
Blind test	Submitted 4-way vote	**43.23**	20.0	4,000

Table 2: Dialect-average error rates (%). The full-development rows use an earlier protocol; the blind-test row is official.

3.3 System combination

The four W2v-BERT models use nested training pools. We name them by approximate training-pool size. *Checkpoint-26K* uses NADI training data and Casablanca; *Checkpoint-29K* adds 2,814 Omnilingual utterances; *Checkpoint-37K* adds 7,474 MoulSot utterances; and *Checkpoint-47K* adds the 10,008 development utterances not reserved for the holdout. They are independent full fine-tunes initialized from the released W2v-BERT 2.0 weights, not one continued run.

The strongest checkpoint, *Checkpoint-47K*, supplies the primary hypothesis. It is trained on the full pool in Table 1; the other models are trained independently on the smaller pools.

The submitted system combines hypotheses with a primary-anchored variant of ROVER (Fiscus, 1997). For each utterance, Levenshtein dynamic programming aligns every secondary word sequence to the primary. We take a majority vote at each primary position, resolve ties in favor of the primary word, and delete a position when deletions have the majority. The submitted four-way system uses *Checkpoint-47K*, *Checkpoint-37K*, *Checkpoint-29K*, and *Checkpoint-26K*; the three-way result omits *Checkpoint-26K*.

4 Results

We report word error rate (WER) and character error rate (CER) after official normalization. Local scores are arithmetic means over the eight dialects, each represented by 100 holdout utterances.

Table 2 separates an early full-development ablation, later local selection, and the official blind test. Every local single-model result uses the same 5-gram decoder. The language model in the early ablation includes the external text described in Section 3.2.

In the early full-development protocol, adding the language model to the same NADI-only acoustic checkpoint lowers WER from 58.25% to 55.37%. These rows should not be compared directly with the later 800-item holdout results.

Folding the released development remainder into training lowers holdout WER from 47.6% for *Checkpoint-37K* to 46.12% for *Checkpoint-47K*.

System combination produces a larger gain. Three-way voting reaches 44.6% WER, and the submitted four-way vote reaches 44.0%. The submitted ensemble is about 2.1 points better than the primary, a 4.6% relative reduction.

The submitted system's WER gain comes with a small local CER increase, from 19.4% to 19.7%. We kept the ensemble because NADI ranks systems by WER and WER improved from one to three to four recognizers.

On the blind test, the four-way system scores 43.23% WER and 20.0% CER. It ranks first, 7.06 absolute WER points ahead of the second-placed system.

The available comparisons locate gains within our pipeline: 2.88 WER points from language-model decoding in the early protocol, about 2.6 points from expanding the later acoustic pool from 26K to 47K, and about 2.1 points from combination. They do not causally explain the 7.06-point cross-team margin because the language-model comparison uses a different protocol and we do not have ablations for the runner-up.

4.1 Dialect analysis

Table 3 breaks down holdout WER for the submitted four-way ensemble. Jordanian and Palestinian speech have the lowest error rates. Mauritanian, Algerian, and Moroccan speech have the highest.

The gap between Jordan and Mauritania is 33.0 points, large enough that the overall average conceals substantial country-level variation. Data coverage and domain mismatch are plausible explanations, but this split cannot establish the cause.

Dialect	WER	Dialect	WER
Jordan	30.7	Yemen	49.1
Palestine	30.2	Morocco	53.4
Egypt	33.0	Algeria	56.2
UAE	36.1	Mauritania	63.7

Table 3: Holdout WER (%) for the submitted four-way ensemble. The 44.0% macro score is computed before dialect-level rounding.

Manual inspection shows whole-word substitutions and deletions in very short clips. Longer clips often contain locally plausible words but accumulate lexical substitutions. Moroccan and Mauritanian examples also expose spelling variation that a character CTC model and fixed n-gram vocabulary do not handle well.

5 Discussion

The submitted combination gains about 2.1 WER points from four models trained on nested acoustic pools. This is a system-engineering result.

The gain is not free. All four submitted recognizers use the same large architecture, so acoustic inference costs roughly four times as much as a single-model run; word voting itself is cheap.

We tried to reduce inference cost through distillation after the submission. Updating the upper half of a teacher-initialized 24-layer student reached 46.09% WER, effectively matching the 46.12% primary rather than compressing it. This result was not used in the submitted system.

Future work should reserve an untouched validation set, tune dialect-specific decoding on separate development folds, and distill on a larger teacher-labeled pool with intermediate-layer matching.

6 Conclusion

Wifaq combines full W2v-BERT 2.0 CTC fine-tuning, 5-gram decoding, and primary-anchored word voting across four nested data variants. The submitted ensemble lowers holdout WER from 46.12% to 44.0% and ranks first in NADI 2026 Subtask 1.1 with 43.23% official WER.

The available ablations attribute gains within our pipeline to language-model decoding, acoustic-data expansion, and system combination. High error rates for Mauritanian, Algerian, and Moroccan speech still call for better dialect coverage, while four-model inference remains expensive.

7 Limitations

The same holdout guided checkpoint, decoder, and ensemble selection, so its gains may be optimistic and small differences may reflect selection noise. We report the number of scored utterances for each result, but the retained hypotheses needed for a valid retrospective paired confidence interval are unavailable. The blind-test ranking supports the final choice but does not remove this local selection bias.

Decoder settings were tuned on Egyptian speech and may be suboptimal for harder dialects. The external corpora differ in genre and transcription process, MoulSot includes automatically generated transcripts, and speaker overlap between Casablanca and NADI remains unknown. Finally, four-model decoding costs roughly four times as much as one recognizer.

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References

Atlasia. 2026. Moulsot: A curated moroccan darija speech dataset and fine-tuned ASR model. Hugging Face dataset. Accessed 2026-09-03.

Loïc Barrault, Yu-An Chung, Mariano Cora Meglioli, David Dale, Ning Dong, Paul-Ambroise Duquenne, et al. 2023. Seamless: Multilingual expressive and streaming speech translation. *arXiv preprint arXiv:2312.05187*.

Yu-An Chung, Yu Zhang, Wei Han, Chung-Cheng Chiu, James Qin, Ruoming Pang, and Yong Wu. 2021. W2v-BERT: Combining contrastive learning and masked language modeling for self-supervised speech pre-training. *arXiv preprint arXiv:2108.06209*.

Jonathan G. Fiscus. 1997. A post-processing system to yield reduced word error rates: Recognizer output voting error reduction (ROVER). In *Proceedings of the 1997 IEEE Workshop on Automatic Speech Recognition and Understanding*.

Alex Graves, Santiago Fernández, Faustino Gomez, and Jürgen Schmidhuber. 2006. Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 369–376.

Kenneth Heafield. 2011. KenLM: Faster and smaller language model queries. In *Proceedings of the Sixth Workshop on Statistical Machine Translation*, pages 187–197. Association for Computational Linguistics.

Gil Keren, Artyom Kozhevnikov, Yen Meng, Christophe Ropers, Matthew Setzler, Skyler Wang, Ife Adebara, Michael Auli, Can Balioglu, et al. 2025. Omnilingual ASR: Open-source multilingual speech recognition for 1600+ languages. *arXiv preprint arXiv:2511.09690*.

Abdelrahman Rezk. 2026. Arabic dialect identification. Hugging Face dataset. Training split used for language-model text; accessed 2026-09-03.

Peter Sullivan, Bashar Talafha, Ahmed Ashraf, Fethi Bougares, Haroun Elleuch, Chiyu Zhang, AbdelRahim Elmadany, Youssef Mohamed, Salima Mdhaffar, Yannick Estève, Mohamed Elhoseiny, Hamzah Luqman, Nizar Habash, and Muhammad Abdul-Mageed. 2026. NADI-2026: The second multidialectal Arabic speech processing shared task. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Bashar Talafha, Karima Kadaoui, Samar Mohamed Magdy, Mariem Habiboullah, Chafei Mohamed Chafei, Ahmed Oumar El-Shangiti, Hiba Zayed, Mohamedou Cheikh Tourad, Rahaf Alhamouri, Rwaa Assi, Aisha Alraeesi, Hour Mohamed, Fakhraddin Alwajih, Abdelrahman Mohamed, Abdellah El Mekki, El Moatez Billah Nagoudi, Benelhadj Djelloul Mama Saadia, Hamzah A. Alsayadi, Walid Al-Dhabyani, and 8 others. 2024. Casablanca: Data and models for multidialectal Arabic speech recognition. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 21745–21758. Association for Computational Linguistics.

Bashar Talafha, Hawau Olamide Toyin, Peter Sullivan, AbdelRahim A. Elmadany, Abdurrahman Juma, Amirbek Djanibekov, Chiyu Zhang, Hamad Alshehhi, Hanan Aldarmaki, Mustafa Jarrar, Nizar Habash, and Muhammad Abdul-Mageed. 2025. NADI 2025: The first multidialectal Arabic speech processing shared task. In *Proceedings of the Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 720–733, Suzhou, China. Association for Computational Linguistics.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, and 3 others. 2020. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45. Association for Computational Linguistics.